

(An Autonomous Institution affiliated to VTU, Belagavi)

Eighth Semester B.E. (Credit System) Degree Examinations

May - June 2023

19ME8X08 - INDUSTRIAL POLLUTION CONTROL

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I

Marks	BT*	CO*	PO*
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|----|----|---|----|-----|---|---|
| 1. | a) | Distinguish between
i) Natural and Man-made air pollutants.
ii) Primary and Secondary air pollutants with examples. | 10 | L*2 | 1 | 1 |
| | b) | Explain sustainable industrial growth with examples. | 10 | L2 | 2 | 1 |
| 2. | a) | Explain in details any five ill effects of Air pollutants giving their possible cause and affected systems. | 10 | L2 | 1 | 1 |
| | b) | Explain with a neat sketch of Wind Rose Diagram (WRD) and also list out the areas of application of meteorology in air pollution control. | 10 | L2 | 2 | 1 |
| 3. | a) | What are the legal steps taken after 1952 great smog and also distinguish between carbon credit and carbon auditing? | 10 | L2 | 1 | 1 |
| | b) | Show the following types of plumes and name the atmospheric lapses that cause them i) Fanning ii) coning iii) Lofting iv) Looping v) Fumigating with neat sketch. | 10 | L2 | 2 | 1 |

Unit – II

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| 4. | a) | Define ESP (Electrostatic precipitator). Explain the working principle of ESP with a neat sketch. | 10 | L2 | 3 | 1 |
| | b) | Explain briefly the causes, effects and control of Hydrocarbons. | 10 | L2 | 4 | 1 |
| 5. | a) | List out the advantages and disadvantages of wet scrubbers. | 10 | L2 | 3 | 1 |
| | b) | List out the sources of smoke and explain how smoke rising from stacks be compared using the Ringelmann Smoke Chart. | 10 | L2 | 4 | 1 |
| 6. | a) | Explain with a neat sketch the working of a bag house filter. What are the suitable materials for the filter fabric at different temperatures? | 10 | L2 | 3 | 1 |
| | b) | Explain any four methods used to control NO _x emission from thermal power plants. | 10 | L2 | 4 | 1 |

Unit – III

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| 7. | a) | Explain Plastic as Pollutant. State its ill effects and explain its control method. | 10 | L2 | 5 | 1 |
| | b) | Explain briefly solid waste management with a neat sketch. | 10 | L2 | 5 | 1 |
| 8. | a) | Explain briefly i). Air (Prevention & Control of Pollution) Act, 1981, ii) Environment Protection Act, iii) Water (Prevention & Control of Pollution) Act, 1974, iv) Wildlife Protection Act, v) Forest (Conservation) Act, 1980. | 10 | L2 | 5 | 1 |
| | b) | Differentiating between noise & sound. List out Odour Control Technologies and explain briefly. | 10 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
